

VANTARA

Smart Supply Chain Intelligence Platform

BUSINESS REQUIREMENT DOCUMENT (BRD)

Revenue, Delivery & Customer Performance Analysis

Prepared by: Seema Kumari | Business Analyst

1. Introduction

Vantara is a supply chain and logistics company that sells products across multiple regions, including USCA, Europe, LATAM, Africa, and Pacific Asia. The company has grown its order volume year over year, but leadership noticed that growth has not always translated into more profit. Some years also show drops in repeat customers and inconsistent delivery performance.

This document explains why this analysis was carried out, what business questions it set out to answer, what data was used, and what was ultimately delivered. It is written for both business stakeholders and technical reviewers, so the project plan and outcomes are clear to anyone reading it, including on GitHub.

2. Why I Chose This Project

I picked this dataset deliberately, not because it was convenient, but because it forces the kind of judgment a real analyst role demands and most portfolio projects avoid:

- It has a genuine hidden problem, not a pre-labeled one. Total order volume rises every year in this data. A shallow read of that trend says “the business is healthy.” Only by pulling revenue per order, order status mix, and delivery performance separately does the real story emerge a sharp Q4 2017 revenue collapse sitting directly underneath rising volume. I wanted a dataset where the obvious headline number is misleading, because that is the situation a business analyst is actually hired to catch.
- It has a wrong answer already sitting in it. An early reference figure put customer loyalty at 76%. Rather than accept a plausible-looking number, the project required recalculating it from first principles in three independent tools (SQL, DAX, Excel) before trusting it. I wanted to practice and demonstrate the discipline of verifying a KPI rather than reporting the first number that shows up in a pivot table.
- It spans the full analyst toolchain end to end. Most portfolio pieces are a single Power BI file with no upstream discipline. I structured this one the way a real engagement runs: a signed-off BRD before any pipeline work, Python for cleaning, PostgreSQL for a proper relational model, SQL for analysis, Excel as an independent cross-check, and Power BI as the executive-facing layer with a Decisions Log documenting every judgment call along the way.
- It rewards going one level deeper, not just reporting the surface finding. Almost every insight in this project (the revenue drop, the late-delivery pattern, the discount-vs-margin relationship) has a first answer and a truer answer underneath it. I wanted a project that punishes stopping at the first plausible explanation, because that habit is the actual difference between a report and an investigation.

In short: I did not choose this project to produce a pretty dashboard. I chose it because the data contains a real diagnostic problem growth masking a collapse, an unverified KPI, an inverted service promise and solving it end to end is the closest a self-directed project can get to simulating an actual analyst engagement.

3. Business Problem

Leadership at Vantara observed the following pattern over 2015–2017:

- Order volume kept increasing, but revenue growth slowed and in some months reversed sharply.
- Delivery times were inconsistent across shipping modes, with a high share of orders arriving late.
- It was unclear how loyal the customer base really was, or which segments deserved more investment.
- A large share of orders sat in unresolved statuses (pending payment, on hold, fraud review), making true revenue hard to pin down.

As Business Analyst, the task was to dig into the order, customer, product, and delivery data, uncover what was really happening, and turn it into clear, practical recommendations leadership could act on.

4. Objective of This Project

The objective was to study Vantara's order, customer, product, and shipping data and answer one core question:

“Where is Vantara losing money and customers, and what should be done about it?”

This was done by building a complete, end-to-end data analysis and reporting solution covering data cleaning, a relational database, SQL analysis, an Excel working model, and an interactive Power BI dashboard backed by documentation explaining every step and decision.

5. Stakeholders

Stakeholder	What They Care About
CEO	Overall company growth and whether the business is heading in the right direction
CFO	Revenue, profit margin, and how much revenue is actually confirmed vs at risk
VP of Operations	Delivery speed, late shipments, and shipping mode reliability
VP of Customer Success	Customer retention, repeat purchase rate, and segment behavior
Marketing Team	Which regions, products, and segments to prioritize for growth

6. Key Business Questions

- Why did revenue drop sharply in late 2017 despite order volume staying high?
- Which product categories and regions bring the most and least profit?
- How reliable is the delivery process, and which shipping modes have the highest late-delivery rates?

- How much of total revenue is actually confirmed (Complete/Closed) versus still pending, on hold, or flagged as fraud risk?
- How loyal are Vantara's customers, and does loyalty differ by segment?
- What specific, data-backed actions can Vantara take to protect profit and improve customer retention?

7. Project Scope

In Scope

- Analysis of historical order, customer, product, shipping, and payment data for full calendar years 2015–2017
- Year-over-year and month-over-month performance comparison
- Profitability analysis by region, category, and customer segment
- Delivery performance analysis by shipping mode and order status
- Customer retention and segment-level behavior analysis
- Building a relational PostgreSQL database, SQL analysis queries, an Excel model, and a Power BI dashboard
- A findings and recommendations report for leadership

Out of Scope

- January 2018 data. The source file contains one partial trailing month (Jan 2018). It was excluded from the analytical scope from the outset, since a single partial month cannot be compared fairly against full calendar years and would distort month-over-month and year-over-year trends. It is retained only in the raw/underlying tables for traceability, not used in any KPI, trend, or dashboard view.
- Real-time / live data tracking (this is a historical analysis, not a live system)
- Direct changes to Vantara's operations, pricing, or systems (this project recommends; operations decides and executes)
- Customer-level personal targeting (analysis stays at segment and aggregate level)

8. Data Source

The analysis uses Vantara's order management data (180,519 order-line records), covering:

- Order details — order date, ship date, sales, profit, discount, quantity, order status
- Customer details — customer ID, segment (Consumer / Corporate / Home Office), city, state, country
- Product details — category, department, price, name
- Shipping details — shipping mode, delivery status, late delivery flag, fulfilment time
- Payment details — payment type (Debit, Transfer, Payment, Cash)
- Location details — city, state, country, market, region

The raw data arrived as a single flat table and was cleaned, validated, and restructured into a proper relational model (one fact table and four dimension tables) before analysis.

Source dataset: DataCo Smart Supply Chain for Big Data Analysis (Kaggle).

<https://www.kaggle.com/datasets/shashwatwork/dataco-smart-supply-chain-for-big-data-analysis>

9. Methodology

- Data Cleaning (Python/Pandas) — fixed data types, removed irrelevant/sensitive columns, standardized text formatting, corrected a small number of malformed records, and excluded the partial Jan 2018 records from all analytical views at load time.
- Data Modeling — split the single table into a star schema: fact_orders plus dim_customers, dim_products, dim_location, and dim_date, connected by primary/foreign keys.
- Database (PostgreSQL) — built the schema, loaded all five tables, and enforced referential integrity.
- SQL Analysis — wrote queries covering revenue trends, profit by category, late delivery rate, repeat customer rate, payment type breakdown, and order status distribution.
- Excel Modeling — built pivot tables mirroring the SQL findings, plus a what-if scenario model estimating the revenue impact of reducing late deliveries.
- Power BI Dashboard — built a three-page interactive dashboard (Overview, Delivery Performance, Customer Insights) on a consistent Vantara brand theme.
- Findings & Recommendations — summarized insights in plain English with clear, actionable recommendations for leadership.

10. Key Performance Indicators (KPIs)

KPI	Why It Matters
Total Revenue	Shows overall financial performance and growth
Profit Margin %	Shows how much of revenue actually becomes profit
Late Delivery Rate	Shows how reliable the shipping process is
Average Fulfilment Time	Shows how fast orders reach customers
Repeat Customer Rate	Shows customer loyalty and retention
Order Status Mix	Shows how much revenue is confirmed vs still at risk

11. Deliverables

- Cleaned, structured dataset (fact_orders, dim_customers, dim_products, dim_location, dim_date)
- PostgreSQL database with full schema and referential integrity
- SQL query file with all analysis queries
- Excel workbook with pivot tables, scenario model, and executive summary
- Power BI dashboard (Overview, Delivery Performance, Customer Insights)
- Assumptions & Decisions Log
- Findings & Recommendations Report

12. Success Criteria

- A clear, data-backed explanation of where revenue and profit came under pressure
- At least five concrete, actionable recommendations for leadership
- A working, easy-to-use Power BI dashboard that leadership can explore independently
- Documentation clear enough that a new analyst could understand the project without additional explanation

End of Document — Vantara Business Requirement Document v2.1 (Final)